

Exercise 1: AFTER Trigger

```
CREATE TABLE EmployeeChanges(  
  ChangeID INT IDENTITY PRIMARY KEY,  
  EmployeeID INT,  
  OldSalary DECIMAL(10,2),  
  NewSalary DECIMAL(10,2),  
  ChangedDate DATETIME DEFAULT GETDATE()  
);
```

```
CREATE TRIGGER trg_AfterSalaryUpdate  
ON Employees  
AFTER UPDATE  
AS  
BEGIN  
  INSERT INTO EmployeeChanges(EmployeeID,OldSalary,NewSalary)  
  SELECT d.EmployeeID,d.Salary,i.Salary  
  FROM deleted d JOIN inserted i  
  ON d.EmployeeID=i.EmployeeID;  
END;
```

EmployeeID	OldSalary	NewSalary
1	5000.00	6500.00

Exercise 2: INSTEAD OF Trigger

```
CREATE TRIGGER trg_PreventDelete  
ON Employees  
INSTEAD OF DELETE  
AS  
BEGIN  
  RAISERROR('Deletion is not allowed',16,1);  
  ROLLBACK TRANSACTION;  
END;
```

Message
Deletion prevented

Exercise 3: LOGON Trigger

```
CREATE TRIGGER trg_MaintenanceLogin  
ON ALL SERVER  
FOR LOGON  
AS  
BEGIN
```

```

IF DATEPART(HOUR,GETDATE()) BETWEEN 2 AND 3
BEGIN
ROLLBACK;
END
END;

```

Login Status
Access denied during maintenance

Exercise 4 & 5: Modify/Delete Trigger

```

-- Modify trigger from SSMS:
ALTER TRIGGER trg_PreventDelete
ON Employees
INSTEAD OF DELETE
AS
PRINT 'Trigger modified';

```

```

-- Delete trigger:
DROP TRIGGER trg_PreventDelete;

```

Exercise 6: Computed Column Trigger

```

ALTER TABLE Employees
ADD AnnualSalary DECIMAL(10,2);

CREATE TRIGGER trg_UpdateAnnualSalary
ON Employees
AFTER UPDATE
AS
BEGIN
UPDATE e
SET AnnualSalary = Salary * 12
FROM Employees e
JOIN inserted i
ON e.EmployeeID=i.EmployeeID;
END;

```

EmployeeID	Salary	AnnualSalary
2	6000.00	72000.00